

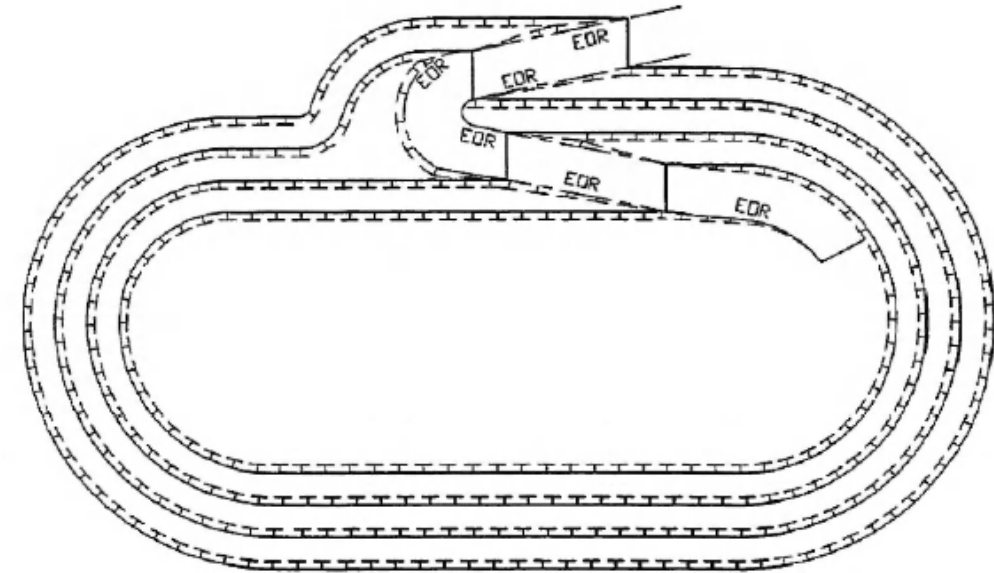
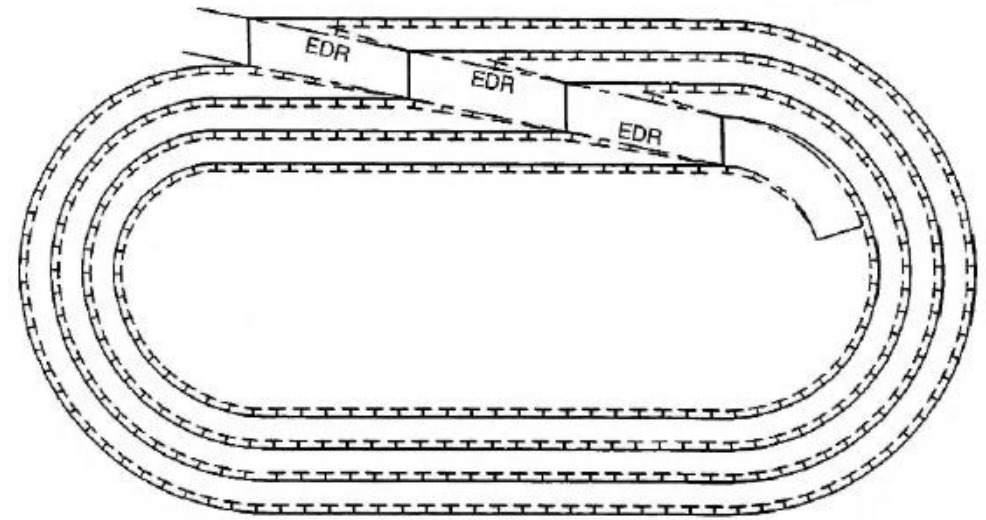
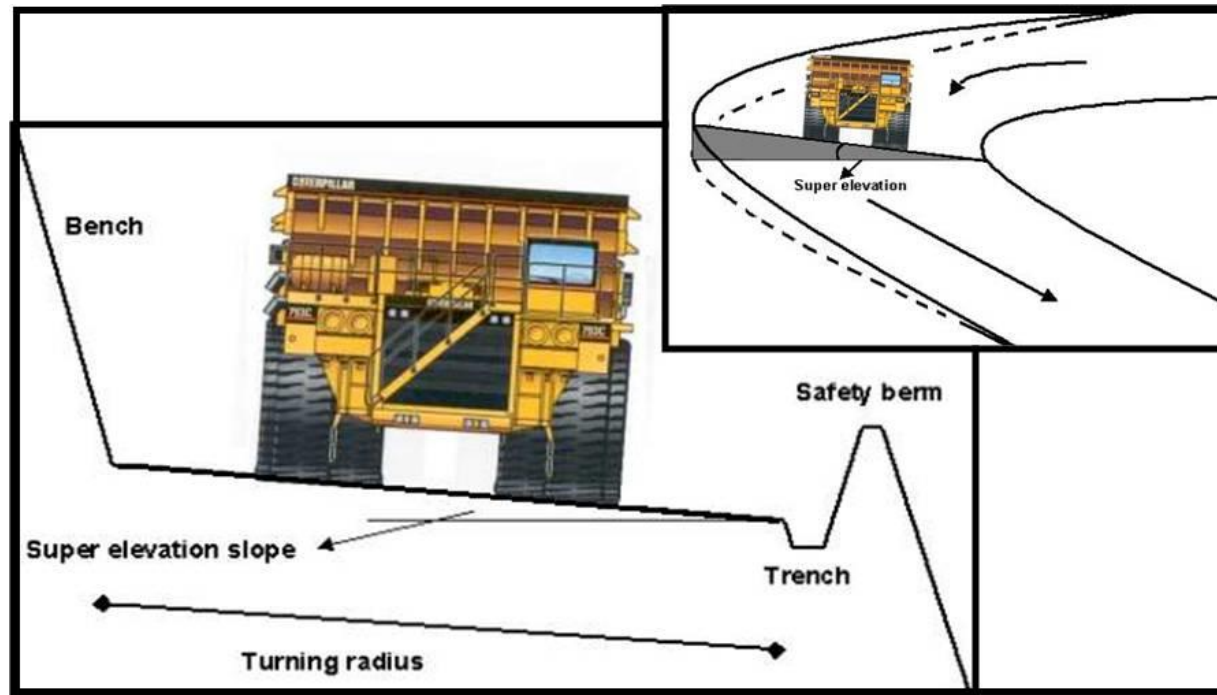
Design of haul roads



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Design of haul roads

- ▶ Geometrical, Structural, Functional and Safety features.



Introduction

- ▶ Major production of coal /iron ore comes from opencast mines through deployment of sophisticated and high capacity dumpers (80t, 120t, and 300t), shovel, dragline, dozer etc. Capital investment on Heavy Earth Moving Machineries (HEMM) is around 70% of the total investment for any open cast project. **Better health of these machineries ensures maximum availability and utilization leading to high production and profit.** To ensure better health of these machineries well designed haul road with continuous improvement in their design parameters is required.
- ▶ A typical surface coal mine has about 3 to 5 km of permanent haul road and various other lumpy roads that are constructed either with overburden material or from locally available material found near to the mine boundary. Some of those materials typically are mudstone, sandstone, gravel, clay etc. Often it is observed that the operating and maintenance cost of dumpers are significantly high in addition to haul road maintenance cost. It results in reduced production, frequent breakdown, accidents, death hazards, low worker motivation etc.
- ▶ These days' opencast mines are planned for greater depths, often beyond industry's current experience, expertise and knowledge. In past 30 years the carrying capacity of hauling equipments e.g. dumpers/trucks has grown from 10t to 170t, 350t being envisioned at places, requiring better haul roads to carry heavy loads. Surface coal mine haul road undergoes more strain due to multiple reasons such as poor surface course, inadequate construction process, poor construction materials, varying load on the surface, improper drainage system, etc.

- ▶ Despite these facts, haul road design until recently has received little attention. The two major functions of haul roads are to promote (1) **efficient transport** and (2) **safety**.

Haul road design factors must ensure:

- ▶ 1. Minimum costs on a net present value basis for the transport of mineral and waste throughout the life of the mine.
- ▶ 2. A minimum of traffic congestion and the maintenance of safe, ready access to the mining operations.
- ▶ 3. The avoidance of areas where slope stability problems could occur.
- ▶ 4. The use of long-life haul roads rather than short-life roads. This reduces haul road overall construction costs and operating costs as well as reducing the demand for haul road construction materials which may not be available in sufficient quantities from the overburden.

Other factors include the locations of mineral preparation plants, stock yards, external waste dumps, environmental constraints, etc. All these factors direct attention to:

- ▶ 1. Haul road layout.
- ▶ 2. Haul road geometry.
- ▶ 3. Haul road construction materials.

HAUL ROAD CLASSIFICATION

- ▶ Haul roads are of three types i.e. permanent, semi-permanent and temporary depending upon the traffic and the nature of its operations.
- ▶ Permanent haul roads are made outside mine boundary to connect approach road to the mine and extend up to the dump yard. These roads are thickest and made of high quality engineering materials and hence construction cost is high. Life of these roads is longer.
- ▶ Semi-permanent haul roads have medium life period, engineered to desired thickness, high quality construction materials, relatively expensive to build, used as main haul roads in mines and dumping yards. The type of roads having lifespan 3 ~ 5 years are often clubbed with permanent haul roads. Materials used are same as permanent haul roads but road thickness is less.
- ▶ Temporary haul road are of short life period, minimum pavement thickness, low quality construction materials, inexpensive to build and used mainly for shovel or dump yard access. They change considerably with the advancement of the mine working face. Typical construction materials are native material from vicinity of the mine.

HAUL ROAD GEOMETRY

► Number of Lanes

In-pit roads are usually constructed for single-lane, uni-directional traffic or two-lane, directional traffic (1) because traffic density may not be high or (2) because of space problems. Haul roads from the pit to external waste dumps, preparation plants, etc., however, may require more than a single lane per direction.

The number of lanes may be determined from the relation,

$$n = \frac{t d_b}{100 v}$$

where n is number of lanes for unidirectional travel, v is vehicle speed in km/h, t is traffic density in vehicles/hr, and db is normal safe distance between trucks in m.

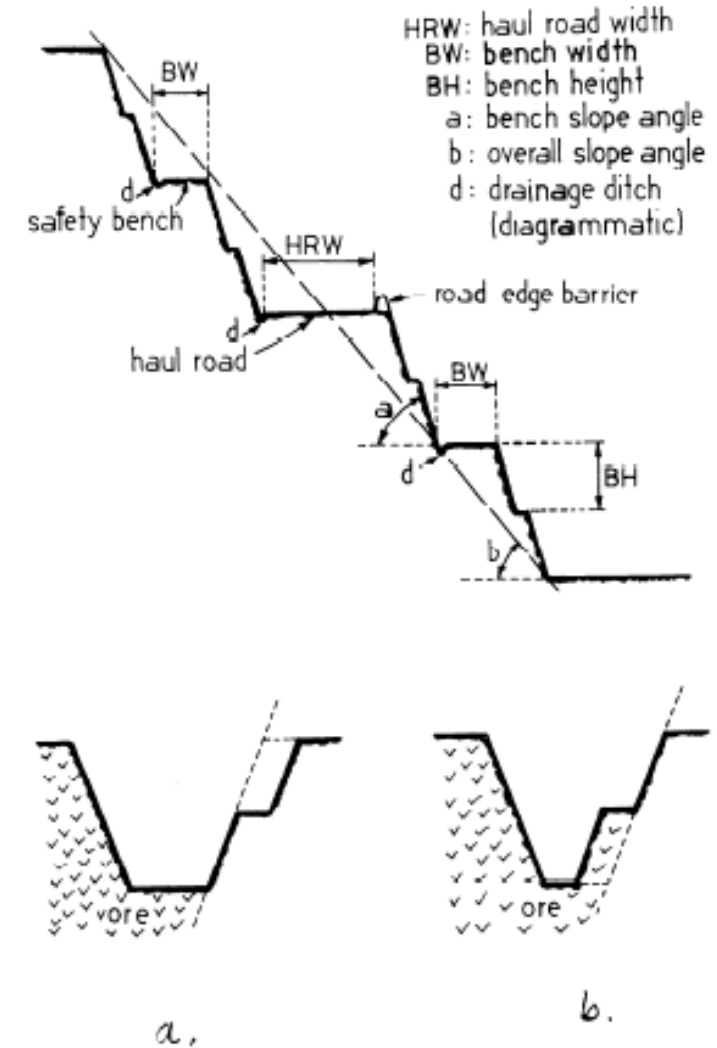


Fig.. Haul road geometric considerations.
 a. Haul road outside ore body. b. Haul road in ore body

Safe Distance Between Trucks

- ▶ The safe distance between trucks depends upon driver reaction time (usually taken as 2.0s), the gradient, and the road surface plus an allowance (usually 5 m). The safe distance can be determined from

$$d_b = \frac{2.0 v}{3.6} + \frac{v^2}{254 (C_t \pm i)} + 5.0$$

where v is vehicle speed in km/h, C_t is coefficient of traction (less than unity), and i is steepest haul road gradient expressed as a fraction.

Road Width

- ▶ The widest vehicles proposed determine the haul road width. For straight, regular grade roads, the rules of thumb given in Table 1 are adequate.
- ▶ For sharp curves, additional width must be included, both on the curve and the tangent to the curve, to cover the front and rear overhangs of the vehicle and the difficulty of negotiating the curve. Minimum percentages of the figures for straight haul roads are given in Table 2.
- ▶ Long tangents to curves assist drivers in negotiating curves.

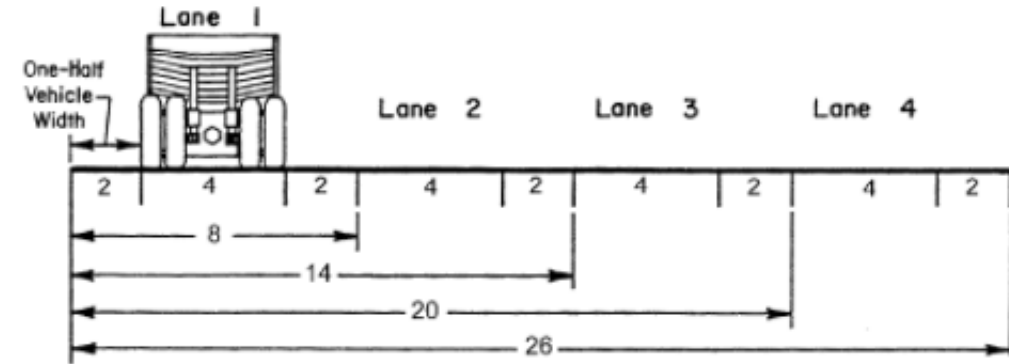


Figure 2-2 Relation between road width and number of traffic lanes with distances in metres for a 4m wide truck (after Monenco 1989).

The minimum width of running surface for the straight sections of single and multi-lane roads can thus be determined from the following expression:

$$W = (1.5L + 0.5)X \quad \text{Equation (5)}$$

Where:

W = width of running surface (m)
L = number of lanes
X = vehicle width (m)

1: Minimum Haul Road Width

No. of Lanes	Factor × Maximum Vehicle Width
1	2.0
2	3.5
3	5.0
4	6.5

2: Additional Allowances

Vehicle	% of Table 13.4.2 (Inside Radius)*		
	20 ft (6 m)	150 ft (45 m)	200 ft (60 m)
Rear-dump and unitized Trucks	125	118	110
Articulated trucks	155	135	115

* Percentages for other radii may be interpolated.

Super Elevation

- ▶ Trucks negotiating tight curves are subjected to an outward centrifugal force, which is opposed by the side friction between the tires and the road surface. Obviously, a good surfacing material is essential on sharp curves, and super elevation of the road surface is normally included in the haul road design. There are practical limitations to super elevation, since trucks driven at slow speeds on sharp curves could (1) overload the tires on the inside of the curve, and (2) in areas of ice, snow, and heavy rain, tend to slide towards the inside of the curve. Table 13.4.4a (or 13.4.4b) shows the super elevation/safe truck speed relationships for practical road construction.
- ▶ Where possible, all the super elevation should be uniformly introduced in the tangent to the curve, the minimum amount being 70% of the total super elevation. Where sharp curves occur at the end of long downhill grades, maximum truck speeds must be restricted to those given in Table 13.4.4 and the maximum permissible travel speed read from Table 13.4.4 to prevent trucks sliding in towards the center of the curve.
- ▶ Tangent lengths vary with truck speeds and total super elevation. Maximum recommended rates of change of super elevation are shown in Table 13.4.5a or 13.4.5b.

Table 13.4.4a. Super Elevation Rates e Expressed in (in./yd)

Truck Speed (mph)	10	15	20	25	30	> 35
Radius (yd)						
5	1.5	1.5	—	—	—	—
10	1.5	1.5	1.5	—	—	—
15	1.5	1.5	1.5	1.8	—	—
25	1.5	1.5	1.5	1.5	2.2	—
30	1.5	1.5	1.5	1.5	1.8	2.2
60	1.5	1.5	1.5	1.5	1.5	1.8
100	1.5	1.5	1.5	1.5	1.5	1.5

Table 13.4.4b. Super Elevation Rates e Expressed in (mm/m)

Truck Speed (km/h)	15	25	35	40	50	> 60
Radius (m)						
15	40	40	—	—	—	—
30	40	40	40	—	—	—
50	40	40	40	50	—	—
75	40	40	40	40	60	—
100	40	40	40	40	50	60
200	40	40	40	40	40	50
300	40	40	40	40	40	40

Table 13.4.5a. Maximum Recommended Rates of Super Elevation Change on Tangents (in./yd)

Truck speed, mi/hr	15	20	25	30	< 40
Super elevation change/100 yd of tangent (in./yd)	9.35	9.00	8.28	7.20	5.76

Table 13.4.5b. Maximum Recommended Rates of Super Elevation Change on Tangents (mm/m)

Truck speed, km/h	5	25	35	40	50	< 60
Super elevation change/100 m of tangent (mm/m)	260	260	250	230	200	160

Gradients

- ▶ Maximum gradients may be statutorily limited to between 8 to 15% (5 to 8.5°) for sustained gradients, but in general when considering the economics of uphill haulage, as well as downhill safety, the optimum gradient for most situations is about 8% (4.5°) but up to 12% (6.8°) for trolley-assist trucks. For safety and drainage reasons, long steep gradients should include 50-m long sections with a maximum gradient of 2% (1°) for every 500 to 600 m of severe gradient.

Safe Sight Distances

Sufficient sight distance must be possible to ensure that a truck can stop when traveling at its operational speed before reaching a hazard. Methods of determining suitable geometries related to safe stopping distances have been advocated using the following criteria (Kaufmann and Ault, 1977).

1. **Vertical curves should provide smooth transitions from one grade to another and provide ample sight distance** for the required braking distance at the operational speed. The sight distance should be taken from the lowest driver's eye height for vehicles in the fleet to a hazardous object 6 in. (150 mm) high as in Fig. 13.4.5a.
2. **Similarly, horizontal curves must be laid out to provide ample sight distance** (Fig. 13.4.5b). This may involve slope reduction at the inside curve.
3. **Where a haul road crosses a public road or a rail track, a safe crossing geometry** is shown in Fig. 13.4.5c. Where vision is obstructed (e.g., by a road located in a trench, trees, vegetation, etc.) 150 ft (50 m) either side of the crossing, a distance to 1300 ft (400 m) back along the public road or rail track must be cleared. Approach gradients should be as flat as practicable.

In some situations, it may not be possible to apply these criteria, and approach speeds must therefore be restricted.

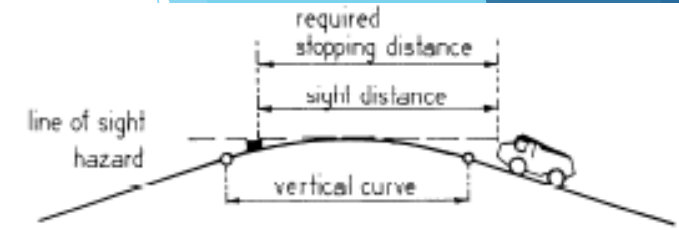


Fig. 13.4.5a. Vertical curve line of sight.

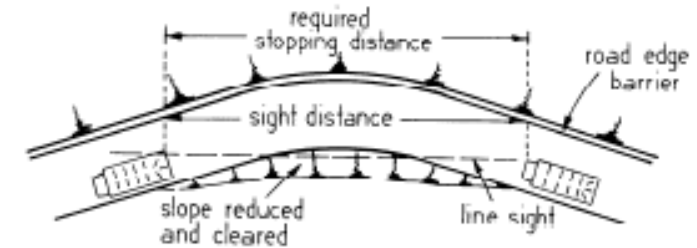
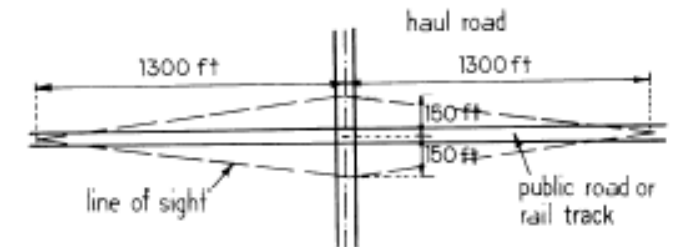


Fig. 13.4.5b. Horizontal curve line of sight.



Conversion factor: 1 ft = 0.3048 m.

Fig. 13.4.5c. Line of sight at route crossing.

Haul Road Signs

- ▶ In general, totally inadequate road signs are used in surface mines since it is often considered that truck drivers become familiar with the route, but the pattern of traffic may be continually changing throughout the life of a mine. Large professionally produced signs with durable surfaces should be installed as needed throughout a mine haul road system. These signs can quickly become obscured by dirt and require periodic cleaning with a high-pressure water jet.

Lighting

Lighting is usually provided at crushers, dump points, etc., to improve efficiency, but the level of illumination must be gradually reduced from an illuminated area to a non-illuminated area to help drivers' eyes to adjust safely to these changes in illumination.



Runaway Precautions

Runaway trucks can be a serious hazard on steep downhill gradients, and safety provisions to guard against these hazards must be provided as part of haul road design. One well-tried method, originating in Australia, is the **location of triangular piles of nonconsolidated fines along the centerline of the haul road**. In the event of brake or retarder failure, the truck driver maneuvers into line with the pile so the truck straddles the pile and the truck is brought to a halt (Fig. 13.4.6a), with only minor damage to the equipment on the underside of the truck. Escape lanes (Fig. 13.4.6b) are a further method available for arresting runaway trucks, but lack of space may prevent their application in many situations. Where switchback haul roads are employed, escape lanes may often be conveniently located at the end of long, **steep grades where the direction of the haul road is reversed**. Particular attention must be paid to the radius of entrance curves, haul road width, super-elevation, wearing materials, arresting materials, etc.

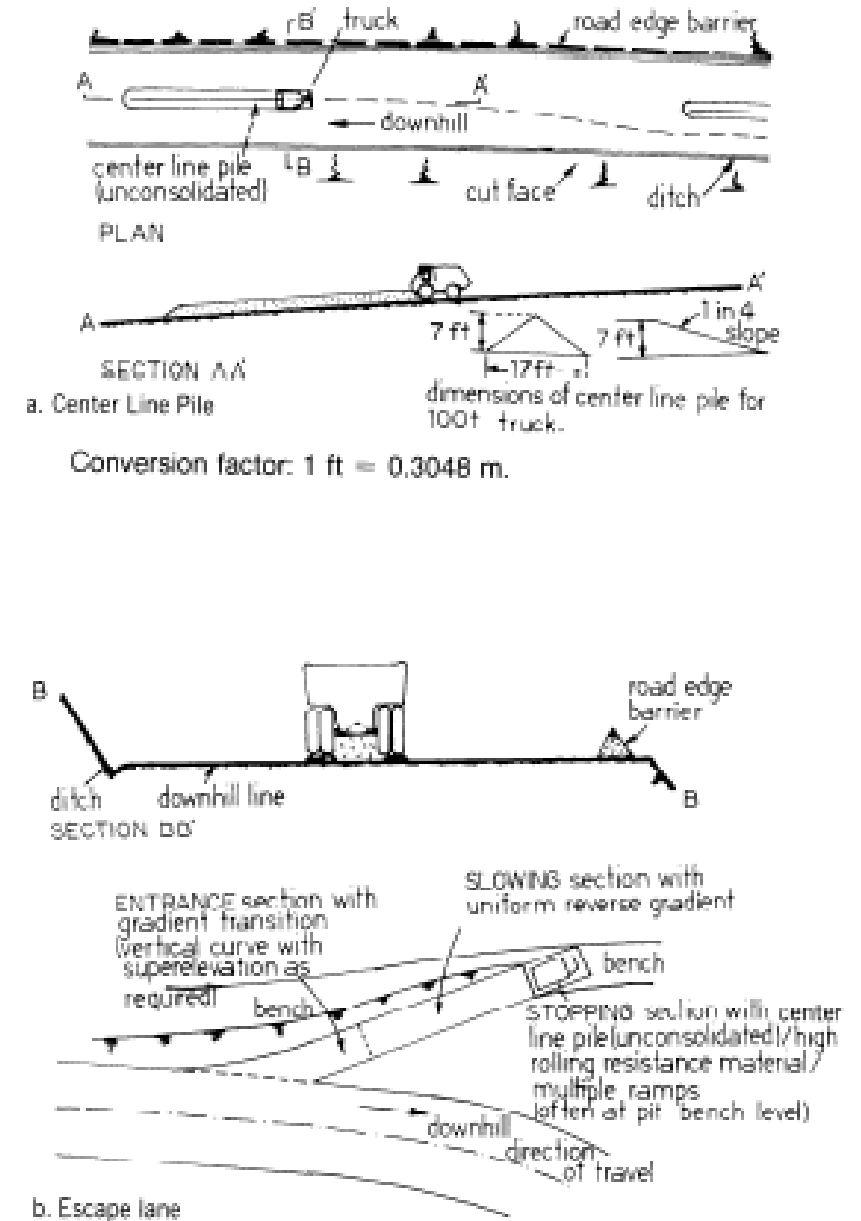


Fig. 13.4.6. Runaway precautions.

Cross Slope

Where possible (e.g., dry situations, short-life roads, etc.), a level surface between road edges is preferable, since this provides more even tire loading and less driver fatigue. **Where heavy rain is experienced, a cross slope is desirable.** Any degree of cross slope must be a compromise that provides adequate drainage without incurring adverse tire loading conditions and driver fatigue.

The normally accepted rate of cross slope is 20 to 40 mm/m depending on conditions.

In conditions of ice, frost, and snow, or on smooth permeable surfaces (e.g., crushed rock) with rock base and sub-base, a 20-mm/m slope is advisable.

For rough surfaces where ice, frost, and snow are not a problem, a cross slope of 40 mm/m should be adopted.

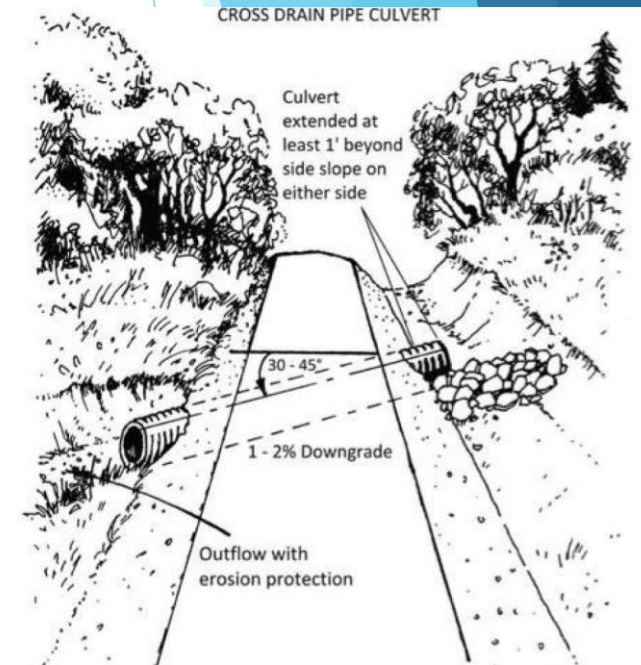
Single- and two-lane haul roads may have all the cross slope in one direction, while on benches, the cross slope should be applied inwards, but three- and four-lane haul roads may have a center high point with the cross slope applied in both directions.

Road edge barriers (rock boulders, windrows of graded material, fines, etc.) should be located between roads and adjacent excavations.



Drains and Culverts

- ▶ Run-off water can create major problems due to washouts, mud slides, and saturation, making provision of drains and culverts essential. The degree of drainage is dependent on rainfall, catchment area, ground conditions, depth of road base, storm water disposal requirements, etc. V drains are generally more easily constructed and maintained, and the following features are desirable:
- ▶ 1. Drains should not be excavated in weak spoil (unless lined with flumes).
- ▶ 2. On benches, the cross slope should drain inward with drains excavated along the toe of the slope above. In cuts, drains should be excavated on both sides of the haul road.
- ▶ 3. Where the haul road is constructed on fill materials, drains should be excavated on each side of the embankment.
- ▶ 4. The ground between the edge of the haul road and the drain should be graded towards the drain and must not be obstructed by debris.
- ▶ 5. The sides of the V drains should have slopes of 4:1 where possible, with 2:1 as a minimum.
- ▶ 6. Drain cross sections must be able to handle the predicted run-offs. The materials in which the drains are excavated can affect the flow rates (e.g., low flow rates are necessary in weak erodable materials).
- ▶ 7. Where possible, the gradient of drains should be restricted to:
 - ▶ < 3% weak materials
 - ▶ 3-5% strong clays, etc.
 - ▶ > 5% crushed rock lining required.
- ▶ 8. Long lengths of down-grade haul roads should be avoided. Sections of flat grade should be included. At these places, the drains should be diverted to the natural drainage system, or a pit sump, through drains or culverts.



DGMS Guide lines

1. HEIGHT AND WIDTH OF HAUL ROAD

- ▶ a. No road shall be of width less than three times plus 5m width of the largest Vehicle playing on road.
- ▶ b. All corner and bends shall be made in such a way that operator of vehicle have clear view of distance of not less than 3 times the breaking distance of largest HEMM working at 40 Km/hour.
- ▶ c. Where it is not possible to ensure a visibility for a distance as mention in (b) there shall be provided with two roads of width not less than 2 times plus 3m of largest vehicle plying on the road with a strong road divider at centre with adequate lighting and reflector along the divider.
- ▶ d. Where any road existing above level of surrounding area it shall be provided with strong parapet wall/embankment of following dimensions.
 - ▶ i. Width at top-not less than 1 m.
 - ▶ ii. Width at bottom-not less than 2.5m
 - ▶ iii. The height not less than diameter of tyre of largest vehicle plying on road
- ▶ It may be noted that just dumping of mud or OB shall not treated as strong parapet wall.
- ▶ e. No road shall have gradient more than 1 in 16. Ramps with 1 in 10 gradients should not be more than 10m at one stretch and permission shall be obtained from Directorate.

DESIGN OF HAUL ROAD PAVEMENT

- ▶ Pavement is a hard, high strength top finished road surface that separates the underlying well-compacted foundation (called sub grade) from the weight of vehicles. There are mainly two types of pavement, **depending upon the strength and type of underlying foundation: (1) Rigid Pavement and (2) Flexible Pavement.** In rigid pavement, most of the load is carried by slab itself and slight load goes to underlying layers. In flexible pavement, load distribution is primarily based on layered system. Structural capacity of rigid pavement depends only on characteristics of concrete slab. Structural capacity of flexible pavement depends on the characteristics of each single layer.
- ▶ Thus, pavement is a structure of superimposed layers of material that is placed on sub-grade. Main function of pavement is to provide friction to the truck and distribute the wheel load to the underlying layers. Pavement deteriorates with time due to interactive effects of traffic load, in-situ material strength, structural thickness and subgrade type. Ideally pavement should be stable, non-yieldable and enables the trucks to move faster with safely and comfortably.

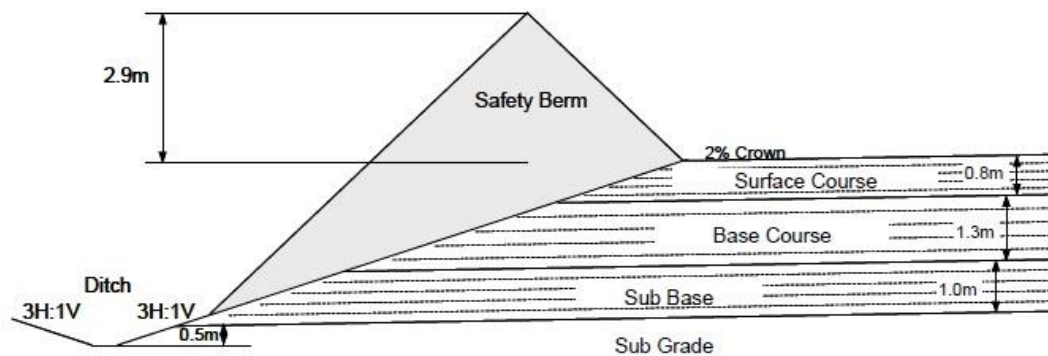
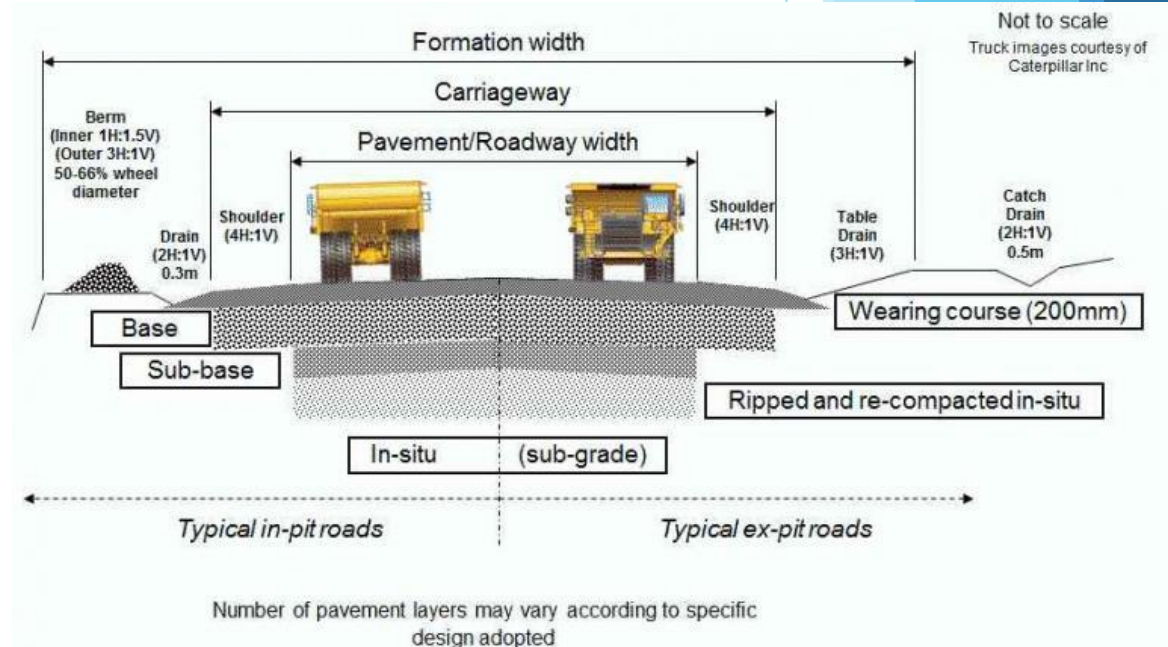
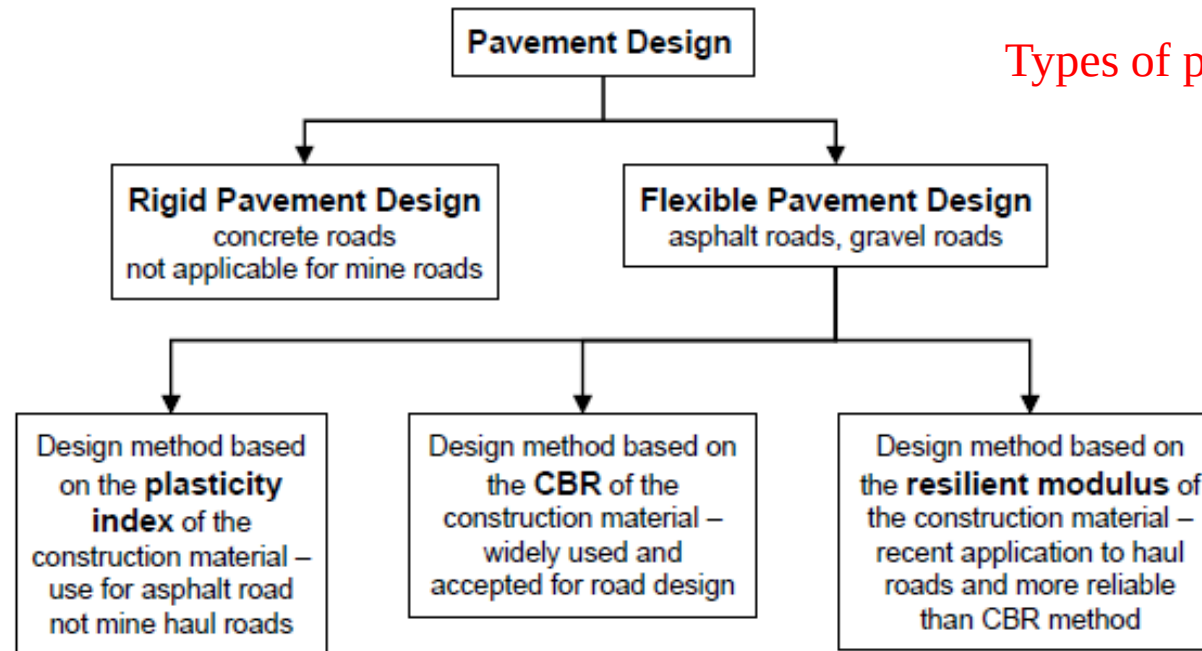


Fig. 2: Typical Haul Road Cross-Section





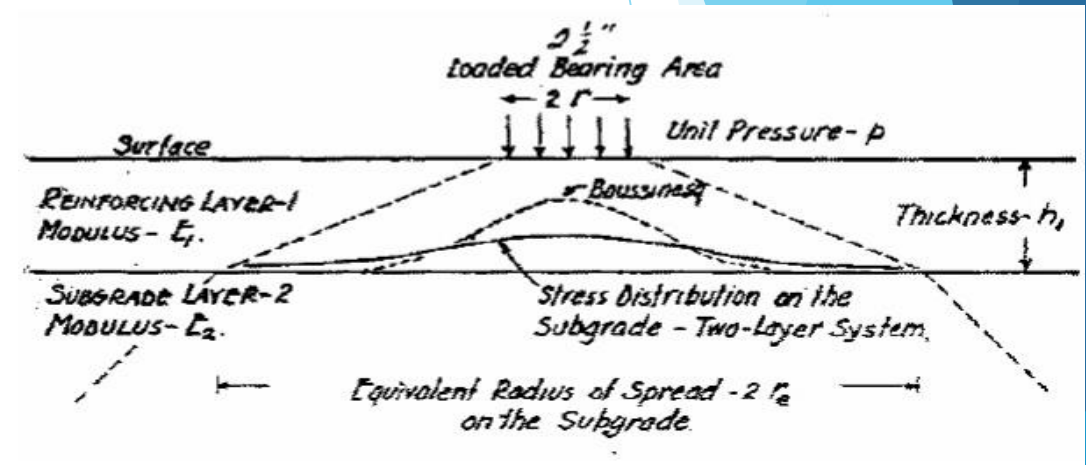
Types of pavement design.

- **Surface course** is top most layer of pavement on which wheels of vehicles are in actual contact. It is generally made of compacted gravel to provide a smooth surface and to resist pressure exerted by the tires. Surface course should have high adhesion, low rolling resistance coefficient, no penetration under load. Earlier bituminous concrete were used as surface course material but now-a-days crushed stone or overburden material are used. Bituminous carpeting is not recommended because (1) at high temperature regions, tire temperature increases further increases due to high tire pressure. This leads to the reduction of tire life because of development of radial cracks in tire; (2) it is getting damaged due to acceleration and braking effect of high capacity dumpers results into formation of large pot holes and ditches; (3) spillage of coal, overburden material and water on damages the road surface.

- ▶ **Base course** is the layer of material which lies immediately below the surface course. It consists of granular material like stone fragments or slag that can be stabilized with binding materials like cement, natural pozzolans etc. The base course is the main source of the structural strength of the road.
- ▶ **Sub-base** is the layer lies between base course and sub-grade. Material used in sub-base are same as base course like laterite, crushed stone, gravel, moorum, natural sand either cemented or untreated. **Apart from providing structural strength to the road, it prevents intrusion of sub-grade soil into the base course, water accumulation and provides working platform for construction equipment.** The base course and sub-base courses are primarily used to improve load supporting capacity by distributing the load.
- ▶ **Sub-grade** is the **naturally occurring surface on which the haul road pavement is constructed.** It may be leveled by excavation or back-filled to provide a suitable surface. The performance of the haul road is affected by the characteristics of the sub-grade. **The loads on the pavement are ultimately received by the sub-grade to be transferred to the earth mass.** It should not be overstresses at anytime i.e. Pressure on top of it should be within permissible limit.

PAVEMENT DESIGN APPROACH

- ▶ Strength-Based Method
- ▶ Strength-based design method uses **shear strength or load-deformation characteristics of the roadbed material**. The strength tests indicate the relative quality of the roadbed materials. Some of the popular strength-based methods have been discussed below.
- ▶ 1. Burmister's method
- ▶ The simplest layered approach in analyzing the pavement is the two layer method introduced by Burmister in year 1943 as shown in Figure. According to this theory, load-settlement characteristics of the two-layer system are influenced by two important ratios i.e. (1) The ratio of the radius of the bearing area to the thickness of the reinforcing or pavement layer and (2) The ratio of the modulus of the subgrade to that of the pavement, for practical design purposes.



Burmister assumed that thickness of each layer is uniform and infinite dimensions in all horizontal directions. Each layer is weightless, homogeneous, isotropic, and linearly elastic. The pavement systems are free from stresses and deformations before application of external loads without any dynamic effects. Presently Burmister's Method is not in use for haul road design.

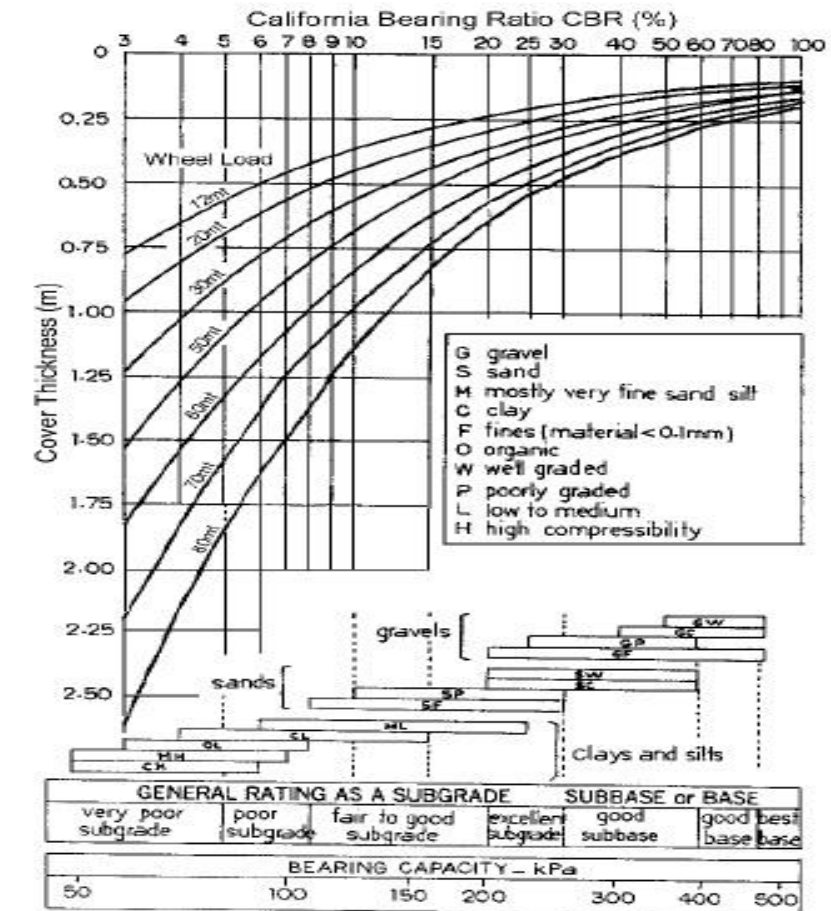
► CBR based design

- California Bearing Ratio (CBR) based design determines the thickness of working surface, base and sub-base based on CBR value determined from laboratory investigation. The standard material for this test is crushed California limestone which has a value of 100. The harder the surface, the higher the CBR value. CBR is defined as follows

- $$CBR = \frac{p}{p_s} \times 100$$

- Where, p = measured pressure for material [N/mm^2], and p_s = pressure to achieve equal penetration on standard material [N/mm^2]

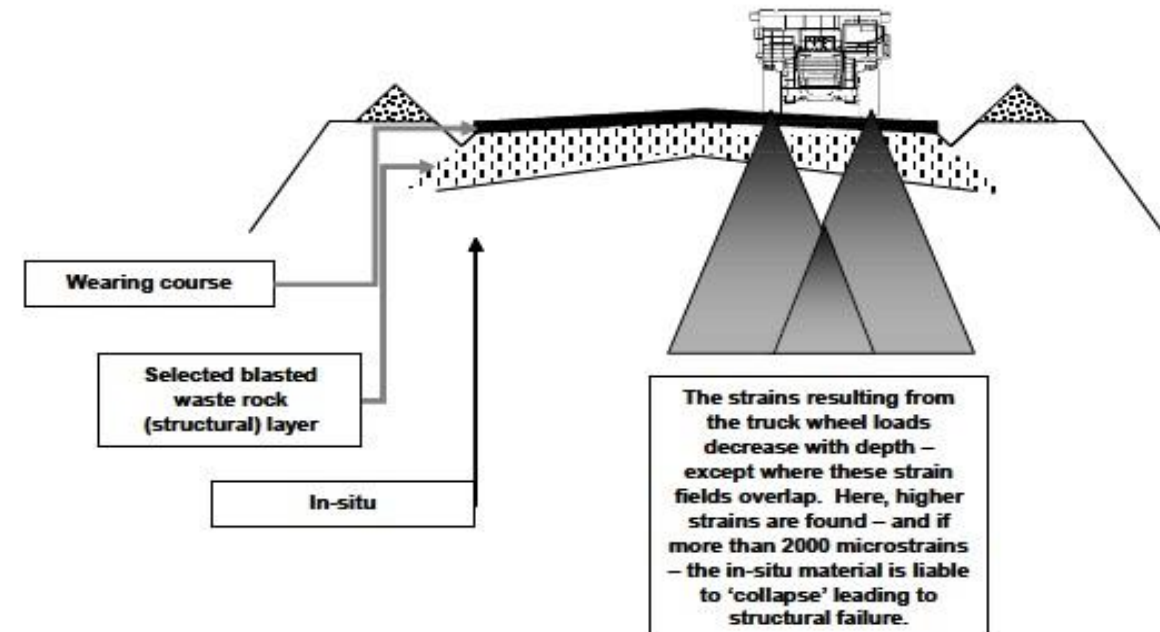
Figure , shows CBR curves given by Atkinson (1992) requires laboratory tests or the assumption of CBR values of sub-grade, and available base or sub-base materials. The CBR curves shows directly the total thickness needed over any sub-grade soil. The total sub-base and base thickness is created by putting down a series of relatively thin layers of correct moisture content. The combined sub-base, base and wearing surface thickness must be sufficiently large so that stresses occurring in the sub-grade will not cause excessive distortion or displacement of the subgrade soil layer. Any sub-grade that is less consolidated than soft rock will require additional material in order to establish a stable base. Finally, the most economic combination of wearing course, base course and sub-base is selected.



- ▶ 3. Mechanistic design
- ▶ A mechanistic design is based on a theoretical linear-elastic multi-layer system model as shown in Figure . A limiting design criterion of vertical compressive strains in the sub-grade is used to assess the structural design adequacy of haul road under the specific loading condition. Vertical compressive strains induced in a road by wheel loads decrease with increasing depth, which permits the use of a gradation of materials. The road as a whole must limit the strains in the sub-grade to an acceptable level and the upper layers must protect the layers below. Applied load, sub-grade strength, pavement structural thickness and layer resilient modulus control the structural performance of a haul road.

Haul Road Category	Typical Description	Range of maximum permissible vertical elastic strains (Micro-strains)	
		Traffic volumes > 100kt/day	Traffic volumes < 100kt/day
Category-I	1. Permanent life-of-mine 2. High traffic volume 3. Main hauling roads and ramps in-and ex-pit 4. Operating life > 20 years	900	1500
Category-II	1. Semi-permanent 2. Medium-to-high traffic volume 3. Ramp roads in-and ex-pit 4. Operating life > 10 years	1500	2000
Category-III	1. Shorter-term 2. Medium-to-low traffic volume 3. Ramp roads in-and ex-pit • Operating life < 5 years (@ > 50kt/day) • Operating life < 10 years (@ < 50kt/day)	2000	2500

Table 1: Haul road classification and associated mechanistic structural design limiting strain criteria (Thompson, 2011)



INDIAN APPROACH OF HAUL ROAD DESIGN (CMPDI & CRRI RECOMMENDED)

- ▶ CMPDI in 1980s designed some haul road cross-sections for various capacity (35t, 50t, 85t and 120t) dumpers. These designs were based on Gray's empirical formula. Gray's formula is an empirical relationship to design rigid pavements. It is a relationship between thickness of pavement (d) in inch; bearing capacity (B) in lbs/sq. inch; (L) radius of circular contact of tire in inch; and static wheel load (W) in lbs. This is only applicable for rigid pavement design:
- ▶ $d = 0.564 \sqrt{\frac{W}{B}} - L$ (Seelye *et al.*, 1948)
- ▶ In 2001 CMPDI and CRRI together developed CBR based design curves and determined designed thickness for various capacities of dumpers (Table 2). The design recommended by CMPDI in 1980 differs subsequently from the design recommended in 2001. Table 3 given below illustrates the major changes.

Table 2: Overall road thickness for different CBR values of subgrade for different capacities of dumpers (CMPDI & CRRI, 2001)

Dumper Capacity	35t					85t					170t				
CBR value of subgrade (%)	3	4	5	6	8	3	4	5	6	8	3	4	5	6	8
Total thickness of pavement (cm)	90	85	75	65	58	135	130	115	100	85	220	190	170	155	130
Thickness of gravel layer (cm)	20	20	20	20	20	30	30	30	30	30	45	45	45	45	45
Thickness of morrum sand layer (cm)	52	47	37	27	20	83	78	63	48	33	145	115	95	80	55

Table 3: Design difference between Gray's formula and CBR method

Parameters	Design of 1980	Design of 2001	Remarks
Carriage way	13.25 m	17.50 m	Width of carriage way increased
Total thickness of pavement crust	825 mm	1150 mm	Increased pavement thickness for same CBR value of subgrade
Construction material	Graded stone aggregates, brick on edge, Bituminous concrete	Sand & crushed rock	Bituminous layer omitted

On comparing both the designs for same load and CBR value of subgrade, the latest design recommends:

- Increased carriage way to avoid accident & provide smooth running of vehicles.
- Increase overall thickness for same CBR value.
- Emphasis on use of locally available overburden material.
- Omission of bituminous layer from surface course.

HAUL ROAD ECONOMICS

The economics of haul road construction are much more complicated than just calculating the cost of road construction. For a true understanding of haul road economics, full life-cycle costs must be considered, and include the following items:

- ▶ road construction costs,
- ▶ road removal costs,
- ▶ impact on fleet productivity and operating cost,
- ▶ differential road maintenance costs,
- ▶ extra fleet operating and maintenance costs ,
- ▶ extra stripping costs, and
- ▶ time value of money.

▶ CONCLUSIONS

- ▶ Construction of mine haul roads differs in many aspects from highways. Wheel loads on haul roads are much higher compared to highways. The problems of design life, continuous change of layout requires a different approach. The haul road designer is required to use continuing ingenuity in layout by the use of cuts, spoil bridges, and ramps, to provide the shortest, least steep and most easily negotiable haul roads. High quality haul roads ensure:
- ▶ Reduced operating costs due to reduced fuel consumption, reduced maintenance cost, and longer tire life.
- ▶ A safer working environment that will improve efficiency.